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Abstract

This coursework has been done with the agenda of improving understanding cryptographic principles and understanding how classical cryptographic methods can be developed using logical and mathematical procedures. The primary issue addressed here is the security of the classical cipher, such as Playfair, which is susceptible to cryptanalytic attacks and does not incorporate any security facilities. This problem was resolved with the development of a new encryption algorithm named EP-36.

The assignments included the pre-processing of the plain text, the generation of digraphs, the creation of the 6x6 Playfair matrix, and the character shifts and XOR operations. Some sample encrypt and decrypt routines have been used for testing the process of encrypting and decrypting the plain. The outcome of this research is evidence that EP-36 is an efficient, simple, and secure cryptographic system. This project is an indication of how classical cryptography can be developed and an effort towards increasing encryption levels.

1. Task 1: Cryptographic Systems

1.1 Security

In computing, security comprises cybersecurity and physical security, which both are used by enterprises to protect against unauthorized access to data centres and other computerized systems. With the fast development of digital technology and online services, information has become a cornerstone to individuals, organizations and even governments. Thus, it has become essential to develop secure information systems for use in the context of contemporary computing infrastructures.

Data protection is about preserving data, whatever its status (storage, processing, storage of a copy) or transmitted across networks. With the lack of a secure mechanism, sensitive information such as personal details can be compromised and bring financial loss, privacy violation or damage company's reputation.

1.2 CIA Triad

CIA Triad is a commonplace model used to guide policies for information security and has become established as the standard upon which organizational I.T. Security programs are based on. It includes as sub-components Confidentiality, Integrity, and Availability — the CIA triad that guarantees the security of information systems.

- Confidentiality

Privacy is the state such that information can be accessed by only authorized parties or systems. It prohibits unauthorised exposure of confidential information. Cryptographic mechanisms like encryption are excellent tools to maintain data privacy by converting the plain text into unreadable information that neither can be read nor write unless one has a valid key.

- Integrity

Integrity is the accuracy and completeness of data. It also prevents data from being modified or corrupted during storage and transmission, moreover it will stop the data being modified.

- Availability

Availability refers to the capability of the user to obtain information or the resources they require whenever they require them in a timely manner. Security systems should be able to defend themselves against attacks such as denial-of-service without failing to provide necessary services. Availability refers to the capability of the user to obtain information or the resources they require whenever they require them in a timely manner. Security systems should be able to defend themselves against attacks such as denial-of-service without failing to provide necessary services.

1.3 Introduction to Cryptography

The term means securing data so only intended parties can see the information. From the Greek, “kryptos” means hidden, and “graphein” means writing. Its main functions are to keep information confidential, ensure that the information is unchanged, and authenticate the identity of the parties involved in the communication. Today, computer systems with cryptography include but are not limited to, secured messaging, online banking, password protection, data privacy, and electronic payment systems. In cryptography, algorithms and logical systems are used to convert plain text to cipher text (Tutorialspoint, 2025).

1.4 Cryptography Terminologies

For a better understanding of the cryptographic systems listed below, please have a look at the below terminologies:

- Plaintext is the content of the communication, or the original message before the encryption.
- Ciphertext is the end output of the encryption process. It is unreadable text unless you go through the process of decryption.
- Encryption is the process by which a plaintext is transformed into ciphertext using a key and an algorithm.
- Decryption is the process of transforming ciphertext into plaintext using a key.
- Key is an integral piece of data, which cryptographic algorithms use to govern the controlling of encryption and decryption processes.
- Cipher is a word that refers solely to the algorithm that is employed for cryptographic purposes to perform either or both the encryption and decryption processes.
- Brute-force attack refers to the attack in which all the possible keys are applied at a certain time, to discover the correct one.

1.5 History of Cryptography

The most reputed form of Cryptography in recent history is the Caesar Cipher which entailed the shifting of alphabets and was used during the time of Julius Caesar for the safeguarding of military messages. However, the history of Cryptography stretches back hundreds and, in some instances, thousands of years. Eventually, the Middle Ages brought about the employment of certain other variation of the cryptography like the repositioning and substitution of certain characters (Tutorialspoint, 2025).

The history of Cryptography, as most understand, took a drastic turn, and underwent innovations as the world entered the mechanical and digital periods like tech advancements with the design of certain mechanical devices for encryption like the Enigma machine. This was a machine widely used during the second world machine.

Finally, the most recent form which is functional Cryptography is the fruit of Contemporary devices which concentrate on the digitalisation of messages and other forms of communication. Contemporary Cryptography is spear headed with the formation of devices like AES, DES, and RSA among several others to protect electronic data and the different forms of virtual interaction. Cryptography is also concerned with the emerging defence challenges surrounding certain modern concepts like IoT, Cloud Computing, and modern mobile devices (Tutorialspoint, 2025).

1.6 Symmetric and Asymmetrical Cryptographic System

Symmetric Cryptography is a form of Cryptography that uses one key for Encrypting and Decrypting the message. Symmetric cryptography uses a single key for both Encrypting and Decrypting the message. The Symmetric cryptography is generally the quickest and most efficient way to Encrypt many GB of data or files using an algorithm such as DES or AES. However, the major drawback of Symmetric cryptography is the difficulty of securely distributing keys.

Asymmetric Cryptography uses two keys for Encrypting and Decrypting a message (or file), the first one being the Public Key and the second one being the Private Key. The security of the key exchange process is enhanced using Asymmetric cryptography; however, the cost of encryption and decryption is greater than that of Symmetric Cryptography. Examples of Asymmetrical Cryptography are RSA and Elliptic Curve. The main applications of Asymmetric Cryptography are secure key exchange and digital signatures.

2. Task 2: Background of Selected Cryptographic Algorithm

2.1 Playfair Cipher

The Playfair Cipher swaps letter pairs instead of single letters, making it a classic code method using shared keys. Created in 1854 by Charles Wheatstone, it got popular thanks to Lord Playfair - hence the name. Unlike simpler codes, this one's tougher to crack, which is why armies and diplomats once relied on it. Because it's straightforward yet more secure than basic symbol replacement, it saw real use in sensitive messages. Instead of swapping single letters like the Caesar method, the Playfair system scrambles two at a time - so letter patterns are harder to spot. Because of this twist, it marked a key step forward in code-making history while also opening doors for later upgrades (Britannica Editors, 2013) (geeksforgeeks, 2025) (Crypto corner, 2023).

2.2 Encryption and Decryption Using Playfair Cipher

Plain Text : HELLO

Key : KEYWORD

Plaintext Preparation : HE LX LO

5×5 matrix

K	E	Y	W	O
R	D	A	B	C
F	G	H	I/J	L
M	N	P	Q	S
T	U	V	X	Z

Table 1 Playfair encryption

Encryption

HE = GY

LX = IZ

LO = SC

Ciphertext = GVIZSC

Decryption

Applying reverse Playfair rules to Ciphertext for Decryption

Plaintext = HELLO

2.3 Advantages and Disadvantages of Playfair Cipher

2.3.1 Advantages

- Scrambles pairs of letters, which makes spotting individual letter patterns harder.
- Easy to grasp, also quick to put into action.
- Needs just one secret key that's shared between parties.
- Faster compared to intricate current encryption methods when handling tiny amounts of info.

2.3.2 Disadvantages

- Easy to crack using letter pair patterns.
- Limited keys when stacked up against today's standards - older systems just don't pack enough variety.
- Fails to deliver strong protection needed today - yet works okay for basic tasks, though not ideal when things get complex.
- Stiff setup limits unpredictability because it holds things in place.

3. Task 3: Development of New Cryptographic Algorithm

3.1 Need for Modification of Playfair Cipher

The classic Playfair Cipher holds up better than basic letter swaps yet stumbles on multiple fronts - its key options are narrow, the grid stays strict at 5 by 5, while deeper math or logic never enters the picture, just spot-based replacements. Because of this, today's code-breaking methods, especially those tracking paired-letter patterns, can crack it without much effort. A fresh take on the Playfair Cipher emerges here - shaped by math-driven tweaks, built for clarity. This twist isn't chasing today's crypto benchmarks; it sidesteps that race entirely. Instead, it walks through how a vintage method can evolve using logic layered with number play.

3.2 Overview of the Algorithm

The name **EP-36 (Enhanced Playfair-36)** is given to the new proposed algorithm. EP-36 draws from the Playfair Cipher, shaping it into a symmetric-key system. Yet instead of multiple rounds, it relies on just one pass for encoding data. To strengthen protection, it widens the range of usable characters beyond traditional limits. With the expansion of Matrix from 5*5 to 6*6

With addition of numbers to the grid. Then, shifted forward by 2 positions. On top of this XOR- based logical transformation is applied to each encrypted character. With converting the alphabet into numeric values as A=0, B=1 Z=25 and 0 = 26 ...9 =35 (geeksforgeeks, 2025).

3.3 Encryption and Decryption Algorithm of EP-36

The steps of the encryption process are as follows:

- The first step of the process is converting the plaintext to uppercase and removing spaces.
- Then, the plaintext must be split into digraphs.
- If a digraph contains repeated characters or if the plaintext length is odd, a filler character (X) is added.
- The 6x6 key matrix must be constructed using the key(A to Z, 0 to 9).
- Then, each digraph is encrypted using the adapted Playfair rules that are meant for the 6x6 matrix.
- Characters of the encrypted digraph is then shifted forward by 2 positions(B=D, Z=1, 9=B).
- Each character of the encrypted digraph is then mapped to their numerical value.
- The numerical value to be worked for the encryption of each character has to be modified through an XOR-based transformation.
- The numerical values obtained are then converted to characters.
- The characters obtained are then collated together to obtain the final ciphertext.

Steps of the decryption process are as follows:

- The ciphertext is split into the digraphs first.
- Then, each character of the digraph has to be mapped to their numerical value.
- XOR to the numerical value using the same key length to reverse the transformation.
- Then the characters are shifted back 2 positions.
- The inverse Playfair rules are then performed for each digraph.
- Any filler characters that the original plaintext would not have been ignored.
- The decryption process can restore the original message as a result of the combination of XOR and the Playfair transformations, which are both fractals.

3.4 Flowchart

Encryption Flowchart

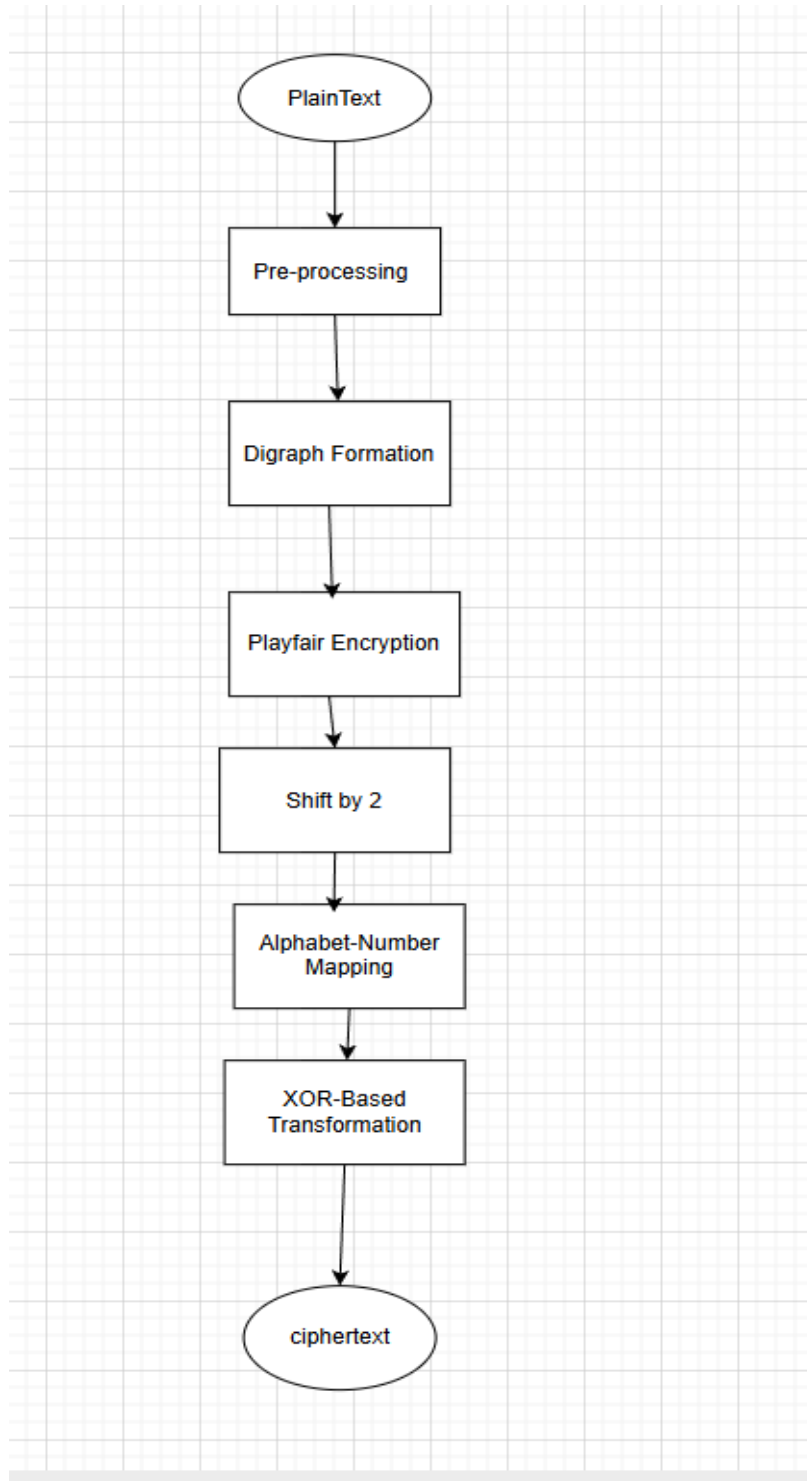
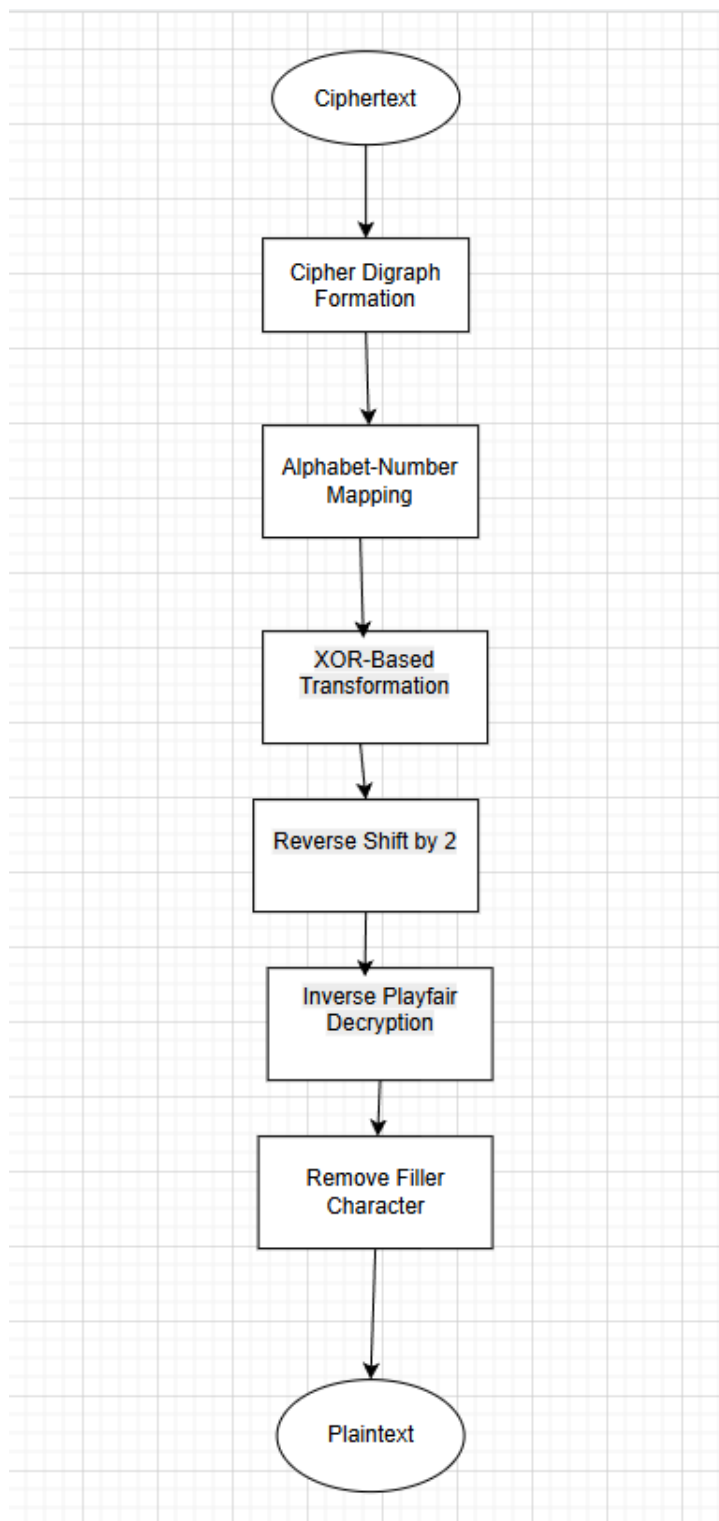


Figure 1 Encryption Flowchart

Decryption Flowchart

*Figure 2 Decryption Flowchart*

4. Task 4: Testing of EP-36 Algorithm

This Section demonstrates the implementation of EP-36 Algorithm by testing multiple cases with different plaintext and various key. Each test case contains successful transformation of plaintext into ciphertext and converting the same back to the original plaintext.

4.1 Test Case 1

Plaintext = ISLINGTON

Key = College

Plaintext in digraph

IS LI NG TO NX

Construction of 6*6 Matrix with key College

Removing Duplicates COLEG

C	O	L	E	G	A
B	D	F	H	I	J
K	M	N	P	Q	R
S	T	U	V	W	X
Y	Z	0	1	2	3
4	5	6	7	8	9

Table 2 Test Case 1

Playfair Encryption

IS = BW

LI = GF

NG = QL

TO = ZD

NX = RU

Shifting Characters forward by 2 positions (Character-level, mode 36)

DYIHSN1FTW

Key length = XOR value = 7 = 000111

Converting letters into Decimal then binary and XOR Transformation. As A=0, B=1
.... Z=25 and 0 = 26 ...9 =35 .

Playfair Text	XOR with 7	Result	Final character
D = 3 = 000011	000111	000100	4 = E
Y = 24 = 011000	000111	011111	31 = 5
I = 8 = 001000	000111	001111	15 = P
H = 7 = 000111	000111	000000	0 = A
S = 18 = 010010	000111	010101	21 = V
N = 13 = 001101	000111	001010	10 = K
1 = 27 = 011011	000111	011100	28 = 2
F = 5 = 000101	000111	000010	2 = C
T = 19 = 010011	000111	010100	20 = U
W = 22 = 010110	000111	010001	17 = R

Table 3 Test Case 1

Final Ciphertext

E5PAVK2CUR

Decryption

Ciphertext = E5PAVK2CUR

Key = college

XOR calculations

Cipher value	XOR 7	Result	Character
E = 000100	000111	000011	3 = D
5 = 011111	000111	011000	24 = Y
P = 001111	000111	001000	8 = I
A = 000000	000111	000111	7 = H
V = 010101	000111	010010	18 =S
K = 001010	000111	001101	13 = N
2 = 011100	000111	011011	27 = 1
C = 000010	000111	000101	5 = F
U = 010100	000111	010011	19 = T
R = 010001	000111	010110	22 = W

Table 4 Test Case 1

Output after XOR

DYHSN1FTW

Reverse Shift by 2

Originalchar = (ShiftedChar - 2) mode 36

= BWGFQLZDRU

Split into Digraph

BW GF QL ZD RU

Inverse Playfair Decryption

BW = IS

GF = LI

QL = NG

ZD = TO

RU = NX

Remove filler character

ISLINGTONX = ISLINGTON

Final Decryption text

ISLINGTON

4.2 Test Case 2

Plaintext = COMPUTER

Key = College

Plaintext in digraph

CO MP UT ER

Construction of 6*6 Matrix with key College

Removing Duplicates COLEG

C	O	L	E	G	A
B	D	F	H	I	J
K	M	N	P	Q	R
S	T	U	V	W	X
Y	Z	0	1	2	3
4	5	6	7	8	9

Table 5 Test Case 2

Playfair Encryption

CO = OL

MP = NQ

UT = VU

ER = AP

Shifting Characters forward by 2 positions (Character-level, mode 36)

QN PS XW CR

XOR Operation

Key length = XOR value = 7 = 000111

Converting letters into Decimal then binary and XOR Transformation. As A=0, B=1 Z=25 and 0 = 26 ...9 =35 .

Playfair Text	XOR with 7	Result	Final character
Q = 16 = 010000	000111	010111	23 = X
N = 13 = 001101	000111	001010	10 = K
P = 15 = 001111	000111	001000	8 = I
S = 18 = 010010	000111	010101	21 = V
X = 23 = 010111	000111	010000	16 = Q
W = 22 = 010110	000111	010001	17 = R
C = 2 = 000010	000111	000101	5 = F
R = 17 = 010001	000111	010110	22 = W

Table 6 Test Case 2

Final Ciphertext

XKIVQRFW

Decryption

Ciphertext = XKIVQRFW

Key = College

XOR calculation

Cipher value	XOR 7	Result	Character
X	000111	010000	16 = Q
K	000111	001101	13 = N
I	000111	001111	15 = P
V	000111	010010	18 = S
Q	000111	010111	23 = X
R	000111	010110	22 = W
F	000111	000010	2 = C
W	000111	010001	17 = R

Table 7 Test Case 2

Reverse Shift by 2

Originalchar = (ShiftChar - 2) mode 36

OLNQVUAP

Into Digraph

OL NQ VU AP

Inverse Playfair

OL = CO

NQ = MP

VU = UT

AP = ER

Final Decryption text

COMPUTER

4.3 Test Case 3

Plaintext = SECURITY

Key = College

Plaintext in digraph

SE CU RI TY

Matrix with key college

C	O	L	E	G	A
B	D	F	H	I	J
K	M	N	P	Q	R
S	T	U	V	W	X
Y	Z	0	1	2	3
4	5	6	7	8	9

Table 8 Test Case 3

Playfair Encryption

SE = VC

CU = LS

RI = QJ

TY = SZ

Shifting by 2 positions

XENUSLU1

XOR Operation

Key Length = XOR value = 7 = 00011

Playfair Text	XOR with 7	Result	Final character
X = 23 = 010111	000111	010000	16 = Q
E = 4 = 000100	000111	000011	3 = D
N = 13 = 001101	000111	001010	10 = K
U = 20 = 010100	000111	010011	19 = T
S = 18 = 010010	000111	010101	21 = V
L = 11 = 001011	000111	001100	12 = M
U = 20 = 010100	000111	010011	19 = T
1 = 27 = 011011	000111	011100	28 = 2

Table 9 Test Case 3

Final Ciphertext

QDKTVMT2

Decryption

Ciphertext = QDKTVMT2

Key = College

XOR Calculation

Cipher value	XOR 7	Result	Character
Q = 010000	000111	010111	23 = X
D = 000011	000111	000100	4 = E
K = 001010	000111	001101	13 = N
T = 010011	000111	010100	20 = U
V = 010101	000111	010010	18 = S
M = 001100	000111	001011	11 = L
T = 010011	000111	010100	20 = U
2 = 011100	000111	011011	27 = 1

Table 10 Test Case 3

XENUSLU1

Reverse shift by 2

VC LS QJ SZ

Inverse Playfair

VC = SE

LS = CU

QJ = RI

SZ = TY

Final text

SECURITY

4.4 Test Case 4

Plaintext = LONDON

Key = Bridge

Into digraph

LO ND ON

Matrix with key BRIDGE

B	R	I	D	G	A
C	E	F	H	J	K
L	M	N	O	P	Q
S	T	U	V	W	X
Y	Z	0	1	2	3
4	5	6	7	8	9

Table 11 Test Case 4

Playfair Encryption

LO = MP

ND = OI

ON = PO

Shift by 2

ORQKRQ

XOR

Key length = XOR value = 6 = 000110

Playfair Text	XOR with 6	Result	Final character
O = 14 = 001110	000110	001000	8 = I
R = 17 = 010001	000110	010111	23 = X
Q = 16 = 010000	000110	010110	22 = W
K = 10 = 001010	000110	001100	12 = M
R = 17 = 010001	000110	010111	23 = X
Q = 16 = 010000	000110	010110	22 = W

Table 12 Test Case 4

Final Ciphertext

IXWMXW

Decryption

Ciphertext = IXWMXM

Key = BRIDGE

XOR calculations

Cipher value	XOR 6	Result	Character
I = 001000	000110	001110	14 = O
X = 010111	000110	010001	17 = R
W = 010110	000110	010000	16 = Q
M = 001100	000110	001010	10 = K
X = 010111	000110	010001	17 = R
W = 010110	000110	010000	16 = Q

Table 13 Test Case 4

ORQKRQ

Reverse Shift by 2

MP OI PO

Inverse Playfair

MP = LO

OI = ND

PO = ON

Final text

LONDON

4.5 Test Case 5

Plaintext = NETWORK

Key = SYSTEM

In digraph

NE TW OR KX

Playfair matrix

S	Y	T	E	M	A
B	C	D	F	G	H
I	J	K	L	N	O
P	Q	R	U	V	W
X	Z	0	1	2	3
4	5	6	7	8	9

Table 14 Test Case 5

Playfair Encryption

NE = LM

TW = AU

OR = KW

KX = IO

Shift by 2

NOCWMYK2

XOR calculation

Key length = 6 = 000110

Playfair Text	XOR with 6	Result	Final character
N = 13 = 001101	000110	001011	11 = L
O = 14 = 001110	000110	001000	8 = I
C = 2 = 000010	000110	000100	4 = E
W = 22 = 010110	000110	010000	16 = Q
M = 12 = 001100	000110	001010	10 = K
Y = 24 = 011000	000110	011110	30 = 4
K = 10 = 001010	000110	001100	12 = M
2 = 28 = 011100	000110	011010	26 = 0

Table 15 Test Case 5

Final Cipher

LIEQK4M0

Decryption

Ciphertext = LIEQK4MO

Key = SYSTEM

XOR Calculation

Cipher value	XOR 6	Result	Character
L = 001011	000110	001101	13 = N
I = 001000	000110	001110	14 = O
E = 000100	000110	000010	2 = C
Q = 010000	000110	010110	22 = W
K = 001010	000110	001100	12 = M
4 = 011110	000110	011000	24 = Y
M = 001100	000110	001010	10 = K
0 = 011010	000110	011100	28 = 2

Table 16 Test Case 5

Reverse shift by 2

LM AU KW IO

Inverse Playfair

LM = NE

AU = TW

KW = OR

IO = KX

Final Text

NETWORK

5. Task 5: Critical Analysis of EP-36

This chapter critically examines the strength and weaknesses inherent in the newly developed cryptographic algorithm, named EP-36, developed and implemented within Task 03. This analysis assesses the efficiency, level of security, usability, and limitations that can be faced by this algorithm on account of its functionalities developed.

5.1 Strengths of EP-36

- **Extended Character Support:**

Unlike traditional Playfair cipher, EP-36 uses a 6×6 matrix supporting alphabets A–Z and digits 0–9. With it, alphanumeric data encryption is possible to enhance usability.

- **Enhanced Security Layers:**

The algorithm makes use of multistage encrypting of the text, This increases confusion and diffusion, a useful layered approach to make cryptanalysis hard.

- **Binary-Level Processing:**

The XOR operation introduces binary computation, adding a touch of modern cryptography to a classical cipher.

- **Reversible Encryption:**

The algorithm is reversible, it allows recovering the original plaintext without data loss, which is of major importance in secure communication.

5.2 Weaknesses of EP-36

- **Fixed Shift Value:**

Using the same shift value of 2 means the system can be broken easily.

- **Single Round Encryption:**

The algorithm is less secure as it only does one round, unlike other more secure multi-round ciphers.

- **Small Key Space:**

The algorithm is easily broken as the key length is small and can be easily brute forced.

- **Not Suitable for High-Security Applications:**

The EP-36 is not an algorithm that can be used for secure systems like banking or military.

5.3 Application Area

- **Educational Uses**

Teaching concepts of cryptography,
Showing hybrid encryption examples (classical + modern).

- **Secure Communication in Class**

Protecting the records of students,
Sharing the information of academics.

- **Light Security Systems**

Low consumption IoT devices,
Basic Authentication Systems.

- **Simulations for Training**

Labs for Cybersecurity,
Tools for practicing encryption.

6. Conclusion

This coursework was based on the design and development of a new encryption algorithm called EP-36, using a modified Playfair cipher technique in conjunction with mathematical and logical operations. In practice, the development of the algorithm entailed a staged approach that covered pre-processing of the plaintext, formation of digraphs, development of a 6x6 matrix, shifting of characters, and performing the XOR operation.

Some decryption and encryption examples were used to test the validity of the proposed method. Close analysis of the method showed advantages and disadvantages of the proposed algorithm and how the traditional method has been made better through advanced binary operations. Overall, this coursework has fulfilled its objectives in designing an encryption system called EP-36 and its assessment as an encryption method. The EP-36 encryption system is appropriate for low-security purposes.

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